PROJECT 4 / MEETING 04 / COMMUNICATION AND CONTROL

Connect a lunar base to three destinations

Task 4.3 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Plan two-way links inside a habitat, from habitat to surface crew and from base to Earth. Sound in air needs a material medium. An electromagnetic signal can cross a vacuum. Draw conversions and a return path.

Mark sender, receiver, and the medium for each required route. Compare indoor, surface, and long-distance links.

Route	Medium / channel	Reason or limitation
Inside base		
Outside crew		
Earth link		

02 Plan it on paper

Choose a channel for each route. Give one reason related to environment, range, or obstruction.

My prediction, chosen feature or design limit

Connect a lunar base to three destinations

Task 4.3 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Create the system diagram and label conversions. Demonstrate an Earth-classroom model of one message path where equipment permits. Ask another team to follow a message from origin to destination. Identify an obstruction, failed component, or missing return path.

Review	Message path or fault	Test result or revision
Trace outward		
Trace return		
One obstruction		

04 Use the reading

Ch. 26 | Shedding Light on Communications | printed pp. 237-244
Ch. 27 | Riding the Waves | printed pp. 245-252
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Defend one channel choice and explain a limit of your classroom test for the lunar setting.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.